

PERCEIVE

$$\hat{u}_t = m_t y_t + (1 - m_t) \Phi(\tilde{u}_t, y_t)$$

WHAT IT DOES

A frozen vision backbone with LoRA adapters, a field decoder and a contrastive text head.

Under occlusion a self supervised estimator fills the cells the satellite did not see.

Occlusion is generated spatially correlated, so a blob covers the front, not stray pixels.

WHAT HELD UP

Losing 60 percent of the observation costs 6.6 of 41 points, and still beats the full state heuristic by a wide margin.

The frozen backbone beats a trained one, t of 3.97, and buys alignment of 0.754.

WHAT DID NOT

Reconstruction is 100 times better on hidden cells, yet the decision was worse in 0 of 9 configurations. Of 4.3 points lost, 3.4 are a representation artefact and 0.86 a real cost under a hard budget.

Bounded by one property of this field: fire covers 1.3 percent of cells, so assuming nothing burns where you cannot see is close to the base rate and costs almost nothing. On a dense field such as flood depth this could invert, which is the next experiment worth running.